Network Dissection of AlexNet and VGG16: an honest reproduction in PyTorch

Interpretable Machine Learning — Group Project
[Group members (names, roll numbers)]

[Submission date] Code: https://github.com/Carbonyx-AI/netdissect-alexnet-vgg

Abstract

We re-implemented Network Dissection [1] from scratch in PyTorch and repeated the paper's experiments for AlexNet and VGG16, using the paper's own Caffe models (AlexNet-Places205, -Hybrid, -Places365, BVLC AlexNet, VGG16-Places365/Hybrid1365, loaded with our own converter) next to torchvision models. **Reproduced:** interpretability is tied to the basis — a random rotation of conv5 of the paper's AlexNet-Places205 removes 84% of the unique detectors (paper: 80%) while the output is unchanged; detectors move from colour/texture to objects with depth; VGG16 > AlexNet; Places > ImageNet for the original models; random seeds converge; detectors emerge during training. **Not reproduced:** batch normalisation does not lower interpretability, removing dropout has no effect, and widening conv5 helps only a little — all three in small-scale training on Imagenette, so these are weak tests. Absolute counts are higher than the paper's, and the size of the rotation effect depends on the model (58–84%). Evaluated with Molnar's criteria [2], single-unit labels have almost no causal importance, while groups of units with the same label do; concepts are captured better by directions than by single units.

1 Introduction

Network Dissection is a *post-hoc, model-specific, global* method (Molnar, *Learned Features*): every unit of a CNN is scored as a segmenter of every concept of a densely labelled dataset, and the number of distinct concepts that some unit detects measures how interpretable a layer is. The paper asks whether interpretable units are an accident of the coordinate system (H1) or a special, axis-aligned basis found by training (H2), and how interpretability depends on layer, architecture, training data and training regime. We implemented the method, checked it against a brute-force IoU, and repeated Sec. 3.1–3.7 for AlexNet and VGG16, the two architectures of our topic. Expectations were written down before running the second round of experiments (`report/EXPECTATIONS.md`).

2 Datasets and models

Broden (`broden1_224`, 63,305 images, labels at 112 px, 1,197 concepts: 584 objects, 234 parts, 468 scenes, 32 materials, 47 textures, 11 colours, exactly the paper's Table 1) is the dataset the method is defined on; dissection needs pixel-level concept labels. The official server was offline, so we used the Internet Archive copy and verified it byte-for-byte (SHA-1). **Imagenette** (10 ImageNet classes) is used for accuracy checks, the fidelity test and the small training runs; **Stanford Action40** (9,532 images, official 4,000/5,532 split, also verified by SHA-1) is the transfer task of the paper's Sec. 3.6. **[note instructor approval of Imagenette / Action40]**

Models. The paper used Caffe models, so we wrote a reader for `.caffemodel`/`.prototxt` files (own protobuf decoder; grouped convolutions, LRN, ceil pooling, BGR mean subtraction). Every converted model was checked before use: BVLC AlexNet reaches 76.7% 1000-way top-1 on Imagenette (torchvision AlexNet 72.0% under the same protocol); on Broden scene images AlexNet-Places205 reaches 62.0% top-1 (205 classes), AlexNet-Places365 46.6%, VGG16-Places365 51.5%. AlexNet-Hybrid is weaker than expected on scenes (38.9% on its 205 scene outputs) and is used with that caveat. Not available: VGG16-Places205 (link dead) and the self-supervised AlexNets (server offline).

Libraries. PyTorch (computation, autograd), torchvision (AlexNet/VGG16 definitions and ImageNet weights), NumPy/pandas (data handling), Pillow (images, label PNGs), SciPy (Schur decomposition, statistical tests), Matplotlib (figures). Dissection, the Caffe reader, rotations, permutation importance, probes and the human check are our code.

3 Method

For each unit k: T_k is the exact top quantile with $P(a_k > T_k) = 0.005$ over all positions of all images; $A_k(x)$ is upsampled bilinearly to 112×112 with each activation at its receptive-field centre and thresholded, $M_k = S_k > T_k$;

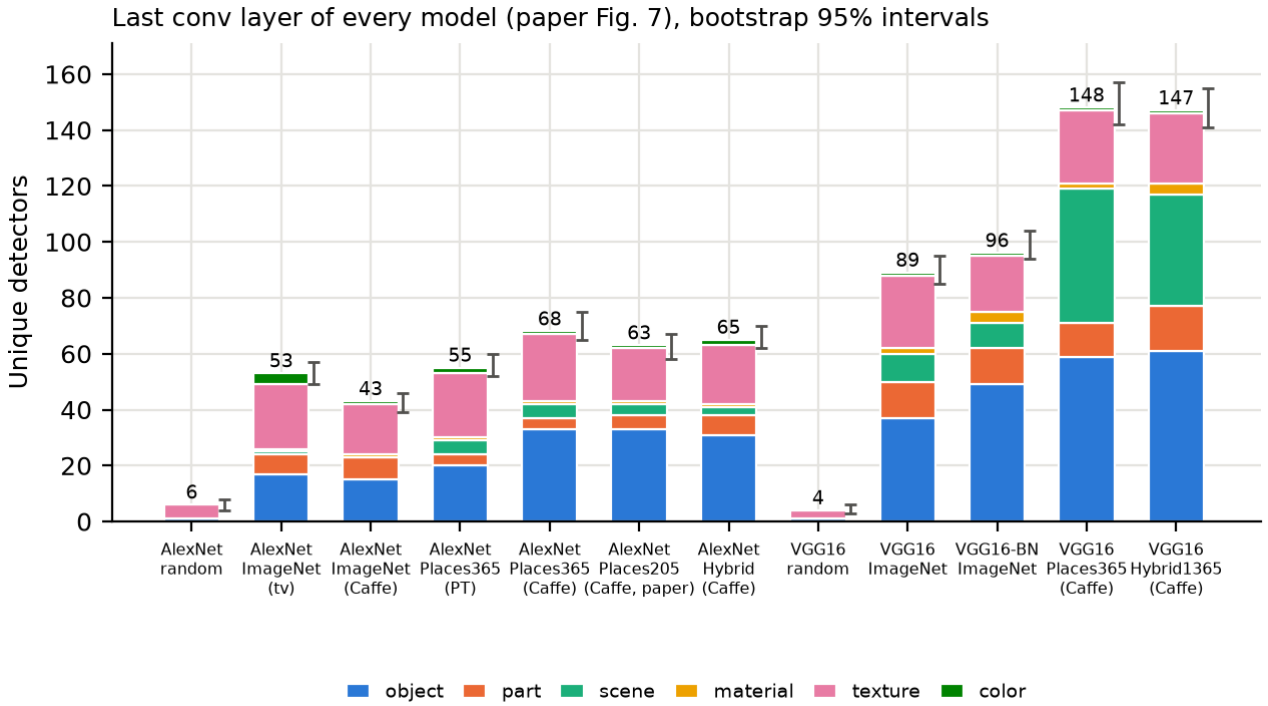


Figure 1: Last conv layer of all 12 models (paper Fig. 7/8). tv = torchvision, PT = PyTorch release, Caffe = original model.

then $\mathrm{IoU}_{k,c} = \sum_x |M_k \cap L_c| / \sum_x |M_k \cup L_c|$ over the images that have a label of c's category; unit k detects its best concept if IoU > 0.04, and a layer's *unique detectors* are the distinct concepts detected. Intersections for all concepts are accumulated per batch with a GPU `index_add` (exact counts), all sums are kept for 50 random image shards (bootstrap), and the quantile is computed exactly with bounded memory (VGG conv1_2 has $3.2 \cdot 10^9$ activations per unit). A random rotation $Q \in SO(K)$ comes from a QR decomposition; Q^α from the real Schur form. Re-running a first-round dissection after the code was refactored gave bit-identical results.

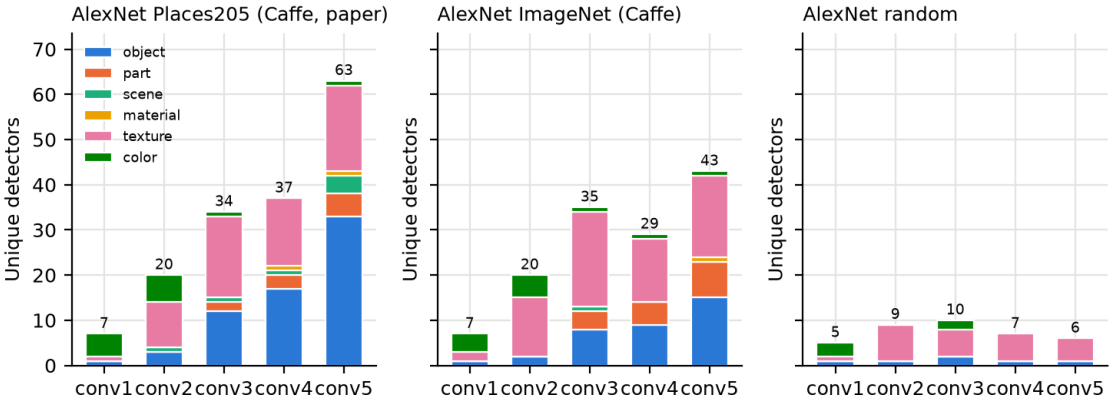
4 Reproduction of the paper

3.4 Architecture and training data (Fig. 1). VGG16 has 89–148 unique detectors, AlexNet 43–68: **reproduced**. With the original Caffe models Places beats ImageNet clearly (AlexNet: Places205 63, Places365 68, Hybrid 65 vs ImageNet 43; VGG16: Places365 148 vs ImageNet 89) and the AlexNet order Places365 > Hybrid > Places205 > ImageNet is the paper's (Fig. 8): **reproduced**. Two caveats: the torchvision ImageNet AlexNet (53) and the PyTorch Places365 release (55) are almost equal, and the PyTorch release has hundreds of units that are zero on >99.5% of positions (the Caffe Places365 model has almost none) — so "Places > ImageNet" holds for the original models, not for every re-implementation. Random networks have 4–6: **reproduced**. Our counts are higher than the paper's (its Fig. 4/5 axes end at 40, we find 63 for the same AlexNet-Places205); we use 224 px instead of 227 px input and cannot otherwise explain the difference.

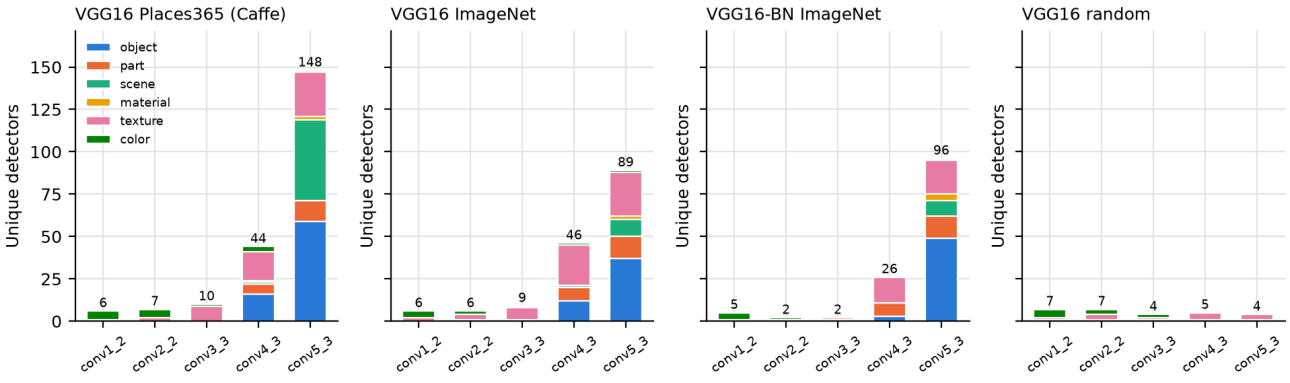
3.3 Concepts by layer (Fig. 2). AlexNet-Places205 rises 7, 20, 34, 37, 63 from conv1 to conv5, colour dominates conv1, textures conv2–conv3, objects conv5 (33 object detectors): **reproduced**. ImageNet and Hybrid AlexNets dip at conv4 (Caffe ImageNet: 35 → 29, although conv3 and conv4 both have 384 units), so the rise is not monotone for every model. VGG16 stays low up to conv3_3 (6–13) and jumps at conv4_3/conv5_3, the same pattern for all trained VGGs.

3.2 Axis-aligned interpretability (Fig. 3a). For the paper's own model, rotating conv5 by a random $Q \in SO(256)$ reduces unique detectors from 63 to 10.0 ± 0.8 (3 rotations), an **84% drop** (paper: 80%), with the collapse between $\alpha = 0.6$ and 0.8 as in the paper. VGG16-Places365 conv5_3: 148 → 30.3 ± 1.2 (80%). The drop is smaller for other models: torchvision AlexNet-ImageNet 58%, Caffe AlexNet-ImageNet 68%, PyTorch AlexNet-Places365 69% (3 seeds each): ImageNet models drop less than Places models in both model families. We also measured the paper's premise: passing $Q^\top Q f(x)$ through the rest of the network leaves the output unchanged (max. logit difference $1.5 \cdot 10^{-5}$), while $Qf(x)$ alone gives chance accuracy. Conclusion: interpretability is a property of the basis, not of the information (H2): **reproduced**.

3.5 Training conditions and 3.7 width (Fig. 4a). The paper trained AlexNet on Places205; we can only train on Imagenette (9.5k images, 60 epochs, 3 seeds per variant), so these are weak, small-scale tests. *Seeds*: 19.0 ± 1.7 unique detectors at equal accuracy (82.5%): **reproduced**. *Iterations*: 4 at initialisation, 4–6 up to iteration 400, 12 at 800, 19 at the end (mean of 3 seeds); object detectors appear after ∼800 iterations: **reproduced** (in shape).



(a) AlexNet, original Caffe models: Places205 (paper's model), ImageNet (BVLC), and a random AlexNet (paper Fig. 5).



(b) VGG16, last conv layer of each block.

Figure 2: Unique detectors per layer.

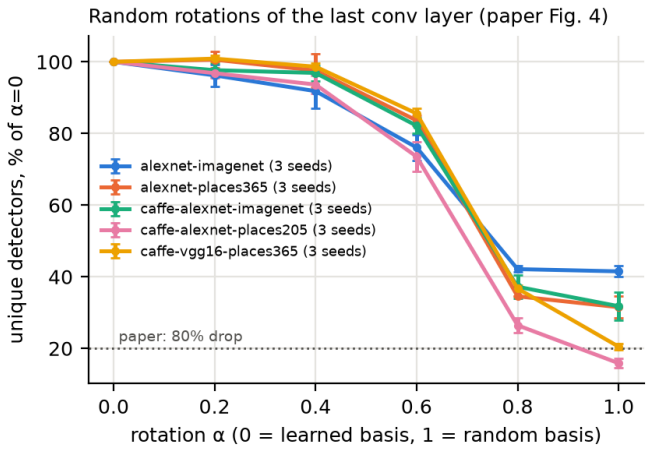
No dropout: 17.3 ± 1.5 (objects 5.0 vs 5.3, textures 9.7 vs 11.3; Welch $p = 0.28$) — the paper's "more textures, fewer objects" is **not reproduced**. *Batch norm*: 23.7 ± 3.8, higher than baseline (not significant, $p = 0.15$) and accuracy 86.3%; pretrained VGG16-BN has fewer detectors than VGG16 in conv3_3/conv4_3 but more in conv5_3 (96 vs 89) — the paper's "BN decreases interpretability" is **not reproduced**. *Width*: replacing the FC layers by global average pooling (GAP) alone already changes the result (256 units: 10.3 unique, 106 detector units, vs 19 / 60 for the baseline), so width must be compared inside the GAP family: $256 \to 768 \to 1024 \to 2048$ units give $10.3, 12.3, 13.3, 15.0$ unique detectors and 106, 327, 426, 884 detector units (3 seeds each). More units give many more detector units but only a few more concepts, and unlike the paper there is no plateau after 768: **partly**.

3.6 Discrimination vs interpretability (Fig. 5a). As in the paper, a linear SVM ($C = 0.001$) is trained on pooled last-conv features for Stanford Action40 and its mean per-class accuracy is compared with the number of unique object detectors. Over the 12 pretrained and random models Spearman $\rho = 0.74$ ($p = 0.006$), the paper's positive relation. Random models are at 7%, AlexNets at 30–35%, VGG16s at 46–60%. But inside each architecture the sign reverses (trained AlexNets $\rho = -0.64$, VGG16s -0.60, not significant): the Places models have more object detectors but transfer *worse* to actions (AlexNet 32% vs 35%). The correlation is driven by architecture and training vs no training: **partly**.

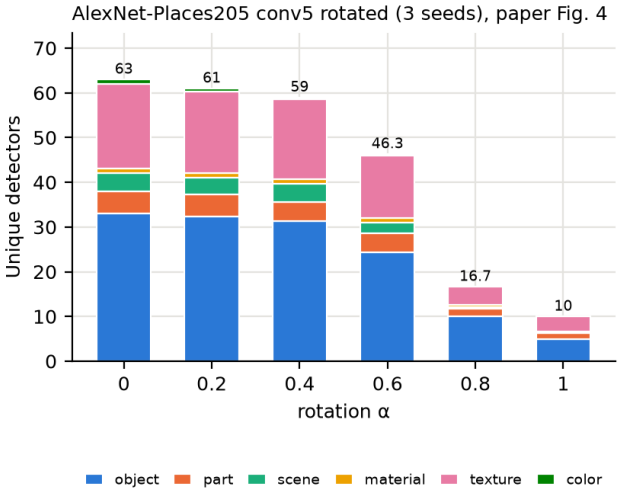
3.1 Human evaluation (Fig. 5b). Paper: raters judged whether the label describes a unit's top images; 71% of conv5 labels of AlexNet-Places205 were accepted. We showed 50 units (8 per layer of AlexNet-Places205, 10 of VGG16-Places365 conv5_3; 10 top images each), half with the true label and half with a decoy of the same category, to [n] rater(s). True labels accepted: [x%]; decoys accepted: [x%]; agreement with the method (Cohen's κ): [x]. With 2–5 true labels per layer, per-layer numbers are noisy.

5 Evaluating the method (Molnar's criteria)

Fidelity. Using permutation feature importance with units as features (Imagenette, AlexNet-ImageNet conv5), labelled units are *not* more important than unlabelled ones (Mann–Whitney $p = 0.80$; VGG16 $p = 0.63$) and IoU does not correlate with importance (Spearman -0.01); no single unit changes accuracy by more than 0.2%. Shuffling all units labelled "dog" together (the one test fixed in advance) costs 24.3 points of 1000-way accuracy on the dog class (13 units; 10,000 random groups: mean 3.4, $p = 10^{-4}$; VGG16: 29.3 points, $p = 0.002$) but under 1 point in 10-way accuracy: the units carry *which breed*, not *dog or not*. For the original Caffe AlexNet-ImageNet the same test is *not* significant (11.8 points vs random mean 4.6, $p = 0.12$; single units: $p = 0.32$): group fidelity of

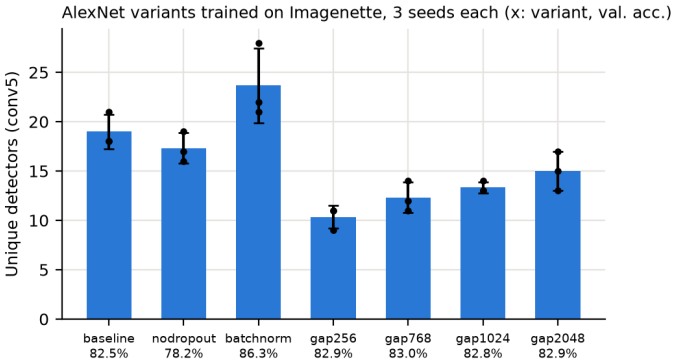


(a) All rotated models (3 seeds each).

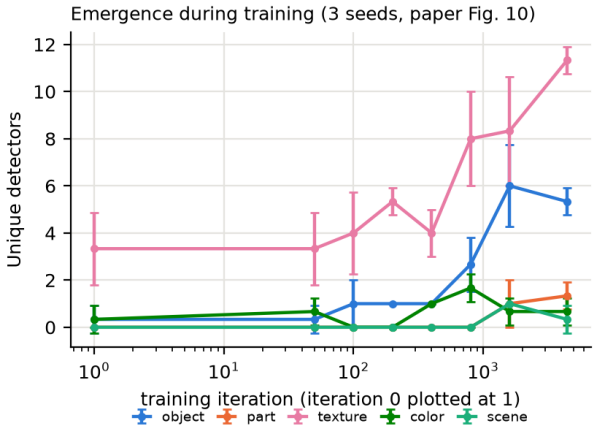


(b) Paper's AlexNet-Places205 (paper Fig. 4).

Figure 3: Random rotations Q^α of the last conv layer.



(a) Variants, 3 seeds each.



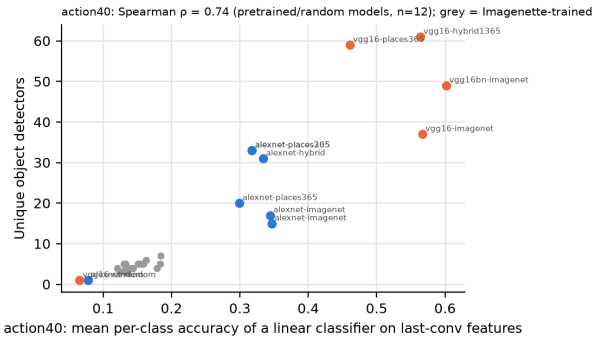
(b) During training (3 seeds).

Figure 4: Small-scale training on Imagenette (paper Fig. 10, 11, 13).

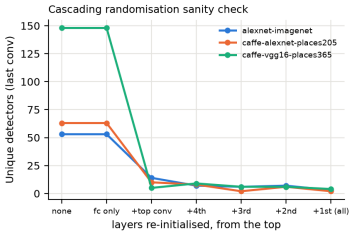
labels depends on the model. **Sanity check** (cascading randomisation). Re-initialising only the classifier changes nothing (labels identical in all three models tested), re-initialising the top conv layer drops the last layer to 5–14 unique detectors (Fig. 5c): the method passes, but it describes what a layer represents, not how the network uses it. **Stability.** 89–94% of last-layer labels of trained models survive bootstrap resampling of images (random models 72–83%); with the 1% quantile 75–88% of labels stay, with 0.1% only 47–63%; Gaussian pixel noise ($\sigma = 0.1$) keeps 65–73%. **Negative activations.** The bottom 0.5% of pre-ReLU activations gives only 10–16 unique concepts (vs 53–148), mostly grid, dotted and chequered textures: the negative tail is not concept-specific. **Units vs directions.** Linear concept directions learned on Broden train images and dissected like units beat the best single unit for 76–100% of concepts in five models and capture 10–15 concepts that no unit detects (Net2Vec [3]). Together with the rotation result: random directions are worse than units, learned directions better — concepts are partly distributed. **Prototypes and criticisms** (Fig. 6). The paper model's "car" unit is clean (96% of its top-100 images contain cars), but most texture units are not: only 3–47% (median 18%) of their top-100 images carry the label, because textures are labelled only on the 5,640 DTD images and IoU is computed only over those images.

Claim of the paper	Verdict	Our evidence
Interpretability is axis-aligned; rotation keeps outputs (3.2)	**reproduced**	−84% for the paper model (paper −80%); 58–84% across models
Colour/texture early, objects late (3.3)	**reproduced**	all trained models; ImageNet AlexNets dip at conv4
VGG > AlexNet; trained ≫ random (3.4)	**reproduced**	89–148 vs 43–68; random 4–6
Places > ImageNet (3.4)	**reproduced**	original models 63–68 vs 43, VGG 148 vs 89; not for torchvision/PyTorch releases
Hybrid between Places and ImageNet (3.4)	**reproduced**	AlexNet 65, VGG 147
Self-supervised < supervised (3.4)	not tested	models unavailable
Seeds converge (3.5)	**reproduced**	19.0 ± 1.7
Detectors emerge during training (3.5)	**reproduced**	$4 \rightarrow 19$, objects after ∼800 it.
No dropout: more texture, fewer objects (3.5)	**not reproduced**	no difference ($p = 0.28$), small scale
Batch norm lowers interpretability (3.5)	**not reproduced**	higher in small models, mixed in VGG16
Discrimination correlates with interpretability (3.6)	**partly**	$\rho = 0.74$ across models, negative within each architecture
Wider layer, more detectors (3.7)	**partly**	more detector units, few more concepts
Orderings stable across IoU thresholds (2.2)	**partly**	8 of 66 model pairs flip, Kendall $\tau = 0.80$
Human agreement 71% at conv5 (3.1)	[]	[]

Table 1: Summary. Of the 12 claims we could test (human evaluation pending), 7 are reproduced, 3 partly and 2 not.



[human check figure]



(a) Action40 vs object detectors (paper Fig. 12). **(b)** Human check (Table 3). **(c)** Cascade.

Figure 5: Discrimination, human evaluation and the randomisation sanity check.

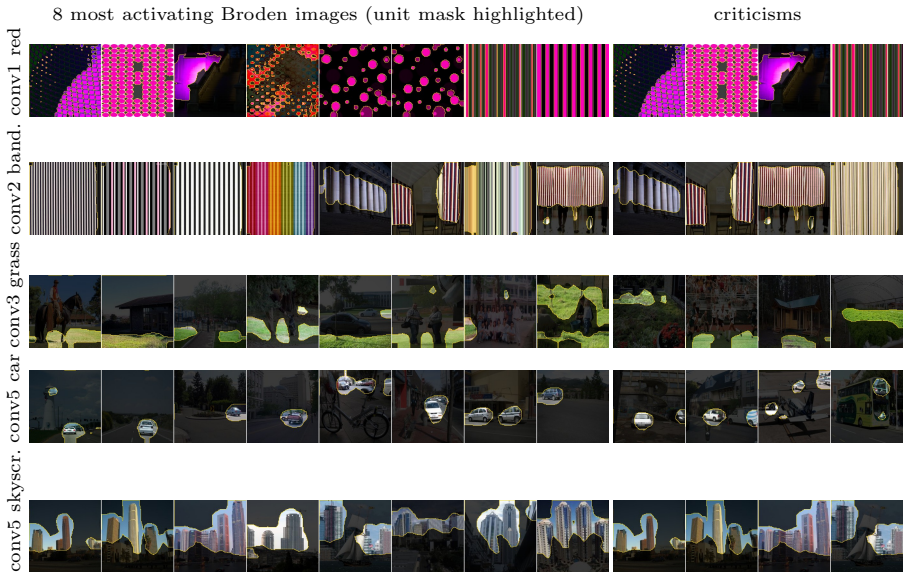


Figure 6: Units of the paper's AlexNet-Places205 (paper Fig. 5 style). Criticisms: top-100 images where the label fits least.

6 Challenges and how we solved them

Challenge	Solution
Broden and Action40 servers offline	Internet Archive copies, parallel ranged download; first Broden file was truncated (index reported the compressed size) — caught by the SHA-1 check, fixed
The paper's models exist only as Caffe files	own protobuf/prototxt reader; every model verified by classification before use; a 5-point accuracy gap traced to the resize protocol, not the converter
RG-encoded label PNGs, up to 5 overlapping maps, image-level scene/texture labels	decoder that merges maps without double counting; scene/texture as full-image masks; label counts checked against Broden's `label.csv`
Exact top quantile over up to $3.2 \cdot 10^9$ values per unit; first version used >200 GB RAM on a model with many all-zero units (tie at 0)	level counting + candidate filtering with explicit tie handling, bounded memory; unit test for ties
In-place ReLU overwrote tensors captured by hooks	non-in-place ReLU, cloned activations, unit test
Upsampling anchored at receptive-field centres	receptive-field start/stride from the layer geometry, `grid_sample`
GAP head confounds the width experiment	added a 256-unit GAP control
Single training runs cannot support conclusions	3 seeds per variant, Welch tests
Our own bugs (wrong weights path, results lost when one model failed)	found from logs, fixed, re-run; every claim checked against the raw files

7 What did not work, bluntly

Batch norm, dropout and width could only be tested on a 9.5k-image dataset; the paper's effects may need the scale of Places205, so our "not reproduced" means "not seen at our scale". Absolute detector counts are at least 1.5× the paper's for the same model (its axes end at 40, we find 63), and we cannot fully explain why. "Places > ImageNet" and the rotation effect size depend on which release of a model is used. Self-supervised models (Fig. 8–9) and VGG16-Places205 could not be obtained. The human check has **[n]** rater(s) and no second group for the paper's "human consistency" row.

8 Conclusion

The central result holds: interpretable units are a property of the learned basis, and the paper's own model loses 84% of its detectors under a random rotation with identical outputs. The comparative results (layer, architecture, training data) also hold for the original models. The training-regime results do not reproduce at our scale. As an explanation method, Network Dissection gives reproducible labels for what units respond to, but single labels say little about what the network relies on — they should be combined with tests of importance, as Molnar recommends.

Declaration. **[State the use of AI tools as required by the course policy.]**

References

[1] D. Bau, B. Zhou, A. Khosla, A. Oliva, A. Torralba. Network Dissection: Quantifying Interpretability of Deep Visual Representations. CVPR 2017.

[2] C. Molnar. *Interpretable Machine Learning.* https://christophm.github.io/interpretable-ml-book, accessed **[date]**.

[3] R. Fong, A. Vedaldi. Net2Vec: Quantifying and Explaining how Concepts are Encoded by Filters in Deep Neural Networks. CVPR 2018.

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[5] B. Yao, X. Jiang, A. Khosla, A. L. Lin, L. Guibas, L. Fei-Fei. Human Action Recognition by Learning Bases of Action Attributes and Parts. ICCV 2011.